25901819

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGPP (HONOURS)

SECOND SEMESTER EXAMINATION

(2025 ADMISSION ONWARDS)

MG2DSCZIM102 – MOLECULES OF LIFE

Duration :1.5 hrs Maximum Marks :50

Students should attempt atleast one question from each course outcome to enhance their overall out come attainability.

Part A

Fill in the blanks

Answer **All** Questions

Each question carries **1** marks

1. Carbohydrates are also known as _____[U] / [CO1]

Answer: Saccharides

2. Myoglobin contains a _____ group, which help in oxygen binding.[U] / [CO3]

Answer: Heme

3. _____ is the principal storage form of energy in adipose tissues.[K] / [CO4]

Answer: Triacylglycerol

4. Phospholipid have a _____ head and two non-polar tail.[U] / [CO5]

Answer: Polar

5. Galactose is the _____ epimer of glucose.[U] / [CO1]

Answer: C-4

6. _____ is a polysaccharide serving as supporting tissue of plants.[K] / [CO1]

Answer: Cellulose

7. The angle of rotation in peptide bond is known as _____ plot.[U] / [CO2]

Answer: Ramachandran

8. _____ can be defined as esters of fatty acid with alcohol.[U] / [CO4]

Answer: Lipids

9. A change in single amino acid in the primary structure due to mutation can lead to disease known as _____.[K] / [CO3]

Answer: Sickle cell anaemia

10. _____ are simple proteins having high affinity to sugar groups.[U] / [CO2]

Answer: Lectins [1 × 10 =10]

Part B

Short Questions

Answer any 10 Questions

Each question carries 2 Marks

11. Describe the importance of chitin.[K] / [CO1]

· Provides structural support for fungal cell walls.
· Forms the hard exoskeleton of arthropods (insects, crustaceans).
· Used in medicine for biodegradable surgical sutures and wound dressings.

12. What is decarboxylation reaction in amino acids?[K] / [CO2]
Decarboxylation is the removal of a carboxyl group from an amino acid as carbondioxide.

13.Define lipoproteins.[U] / [CO5]
Lipoproteins are complexes of proteins and lipids that transport lipids in blood.Lipoproteins transport cholesterol and triglycerides. Examples are HDL and LDL

14. Write short note on lipoprotein.[U] / [CO2]
Lipoproteins are complexes of proteins and lipids that transport lipids in blood.Lipoproteins transport cholesterol and triglycerides. Examples are HDL and LDL.

15. Outline chemical structure of nucleic acids.[U] / [CO5]
Nucleic acids are made of polymers of nucleotides containing sugar, phosphate and nitrogen base.

16. List out the main types of disaccharides found in nature.[K] / [CO1]
Sucrose, lactose and maltose.
· **Sucrose**: Glucose + Fructose (table sugar).
· **Lactose**: Glucose + Galactose (milk sugar).
· **Maltose**: Glucose + Glucose (malt sugar).

17 .Name two factors that can cause protein denaturation.[U] / [CO2]
High temperature and extreme pH.

18. Explain the relationship between iodine value and double bond.[U] / [CO4]
Higher iodine value indicates greater number of double bonds.

19. Write a note on formation of Osazone with phenyl hydrazine.[U] / [CO1]
Reducing sugars react with phenyl hydrazine forming yellow crystalline osazones.

20. What are the complementary base pairs in DNA according to Watson and Crick?
[K] / **[CO5]**
Adenine pairs with thymine via two hydrogen bond and guanine pairs with cytosine via three hydrogen bond.

21. Distinguish between amylose and amylopectin.[U] / [CO1]
· **Amylose**: Unbranched, linear chain with alpha-1,4-glycosidic bonds.
· **Amylopectin**: Branched chain with alpha-1,4 and alpha-1,6-glycosidic bonds.

22. Write any two properties of zwitter ion.[K] / [CO2]
They possess both positive and negative charges and show amphoteric nature.

[2x10 = 20]

Part –C
Short Essay Type Questions
Answer 5 Questions

Each question carriers 4 marks

23. Write an essay on denaturation and renaturation of DNA.[K] / [CO5]
(Definition -2 mark,Factors-2 mark,)
Denaturation separates DNA strands by breaking hydrogen bonds. Renaturation rejoins complementary strands restoring double helix.

24. Differentiate between epimer and anomer.[U] / [CO1]
(Definition-2 mark,Structural representation-2 mark,)
Epimers differ at one chiral carbon other than anomeric carbon. Anomers differ at anomeric carbon.

25. Illustrate the structure of acidic amino acids.
(Structural illustration -2 mark each,Value liberally) [A][CO5]
Acidic amino acids like aspartic acid and glutamic acid contain extra carboxyl groups.

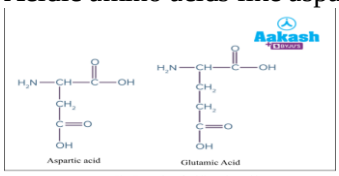


Fig: Examples of acidic amino acid

26. Compare and contrast fructose with glucose including their structure.[U] / [CO1]
(Structural difference-2 Mark ,Comparison-2 Mark)
Glucose is an aldohexose while fructose is a ketohexose. Both are monosaccharides with different structures.

27. Define the quaternary structure of proteins and explain how it differs from tertiary structure.[K] / [CO3]
(Definition-1 mark,Describing functionality- 3mark)
Quaternary structure is the highest level of protein organization. It describes how multiple individual polypeptide chains (subunits) join together to form a single, functional protein complex. These subunits are held together by the same non-covalent forces found in tertiary structures, such as hydrogen bonds, ionic bonds, and hydrophobic interactions.

28. Compare and contrast saturated fatty acids from unsaturated fatty acids.[K] / [CO5]
(Saturated fatty acid with one example-2 mark,Unsaturated fatty acid with example-2 mark)
Saturated fatty acids lack double bonds whereas unsaturated fatty acids contain one or more double bonds.
Eg:Palmitic acid,Butyric acid-Saturated fatty acid
Oleic acid,Linoleic acid-Unsaturated fatty acid

29. Discuss the significance of heparin.[K] / [CO1]
(Any 4 significance-1 mark each)
Heparin is an anticoagulant used to prevent blood clotting during medical treatments.It prevents blood from clotting during open-heart surgery and dialysis when blood passes through machines. Used in hospitals to treat deep vein thrombosis (DVT), pulmonary embolism (PE), and acute coronary syndrome

[4 ×5 =20]